Sanaul Haque

Pierre_Phoella_G._Ilagan_Proposal

Assignment 1066

Document Details

Submission ID
trn:oid:::3117:636713403

Submission Date
Oct 9, 2026, 11:57 PM GMT+6

Download Date
Oct 9, 2026, 11:59 PM GMT+6

File Name
Pierre_Phoella_G._Ilagan_Proposal.pdf

File Size
467.6 KB

7 Pages

2,583 Words

14,921 Characters

32% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **27** Not Cited or Quoted 26%
  Matches with neither in-text citation nor quotation marks
- **3** Missing Quotations 1%
  Matches that are still very similar to source material
- **8** Missing Citation 6%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

26% Internet sources
25% Publications
27% Submitted works (Student Papers)

Match Groups

- **27** Not Cited or Quoted 26%
  Matches with neither in-text citation nor quotation marks
- **3** Missing Quotations 1%
  Matches that are still very similar to source material
- **8** Missing Citation 6%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

26% Internet sources
25% Publications
27% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	panel.bidgecongress.org 3%
2	Internet	assets-eu.researchsquare.com 2%
3	Publication	Asmail A.M. Abdalkarem, Ahmad Fazlizan, Najm Addin Al-Khawlani, Wan Khairul ... 2%
4	Publication	Md. Anonno Habib Akash, Md. Shameem Hossain, Md. Monir Hossain, Md. Nasirul... 2%
5	Student papers	University of Nottingham on 2025-05-06 1%
6	Publication	Cezara-Liliana Raţ, Adela Berdie, Raluca Rob, Catalin Ichim-Burlacu. "Prospects of ... 1%
7	Student papers	International University of Sarajevo on 2025-06-03 1%
8	Internet	khazna.ku.ac.ae 1%
9	Student papers	Glyndwr University on 2025-01-16 1%
10	Student papers	University of Southampton on 2022-05-05 1%

11	Internet	
www.e3s-conferences.org		1%
12	Student papers	
University College London on 2024-09-11		1%
13	Student papers	
University of Aberdeen on 2025-08-15		1%
14	Student papers	
University of New South Wales on 2025-03-30		1%
15	Student papers	
AlHussein Technical University on 2024-01-31		1%
16	Student papers	
University of South Africa (UNISA) on 2023-11-12		<1%
17	Internet	
kuet.ac.bd		<1%
18	Internet	
www.diventures.com		<1%
19	Internet	
pure.tue.nl		<1%
20	Publication	
Amr Khedr. "Bio-mimicry in the aerodynamics of small horizontal axis wind turbi...		<1%
21	Internet	
assets.ctfassets.net		<1%
22	Student papers	
University of Waikato on 2025-08-10		<1%
23	Publication	
Amiyo Krishnamurtti Saha, Mahir Anjum, Zabir Mahmud, Md Hasanuzzaman. "Wi...		<1%
24	Student papers	
De La Salle University on 2022-05-21		<1%

25 Student papers

Khalifa University of Science Technology and Research on 2022-03-11 <1%

26 Internet

ouci.dntb.gov.ua <1%

27 Internet

reutersinstitute.politics.ox.ac.uk <1%

28 Publication

Britney Arce, Mary Lynn Davis-Ajami, Bailey Ortyl, Kathryn Hartlieb et al. "'eatNpl... <1%

29 Internet

internationaljournalssrp.org <1%

30 Internet

rjpn.org <1%

31 Internet

www.irjet.net <1%

32 Student papers

Far Eastern University on 2025-06-27 <1%

33 Publication

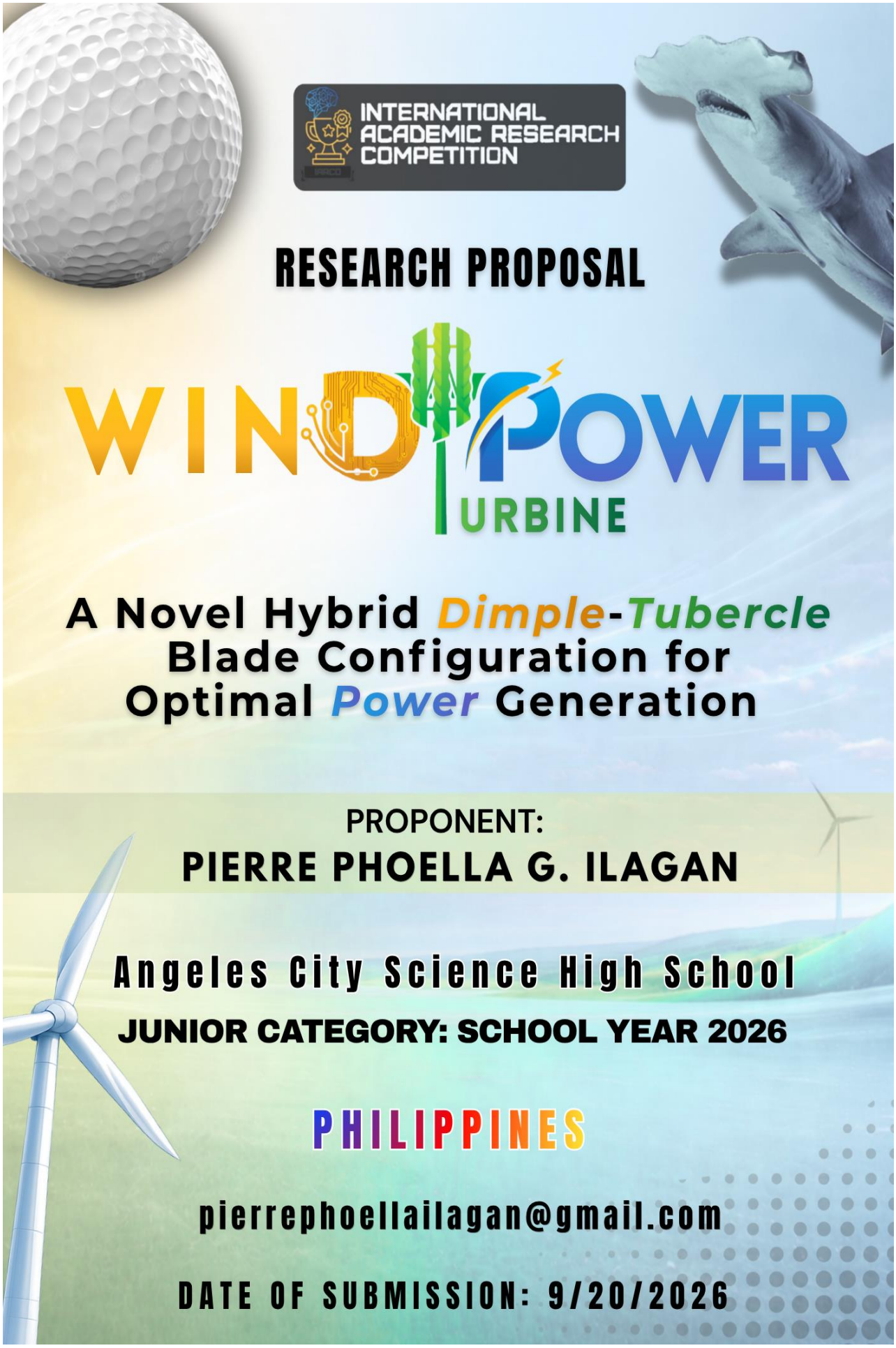
Saepul Rahmat, Erna Alimudin, Riyani Prima Dewi, Nike Sartika, Hendi Purnata, A... <1%

34 Internet

www.schs.state.nc.us <1%

35 Student papers

University of Wales Swansea on 2026-02-25 <1%


INTERNATIONAL ACADEMIC RESEARCH COMPETITION
RESEARCH PROPOSAL
WIND POWER TURBINE
A Novel Hybrid Dimple-Tubercle Blade Configuration for Optimal Power Generation
PROPONENT:
PIERRE PHOELLA G. ILAGAN
Angeles City Science High School
JUNIOR CATEGORY: SCHOOL YEAR 2026
PHILIPPINES
pierrephoellailagan@gmail.com
DATE OF SUBMISSION: 9/20/2026

Renewable energy sources are essential in lessening world's reliance on fossil fuels, accounting for nearly 60% of global power generation [1]. 35% alone being contributed by coal-based power [2]. Currently dependent on fossil fuels, the Philippines is powered by 22% renewable sources, targeting increase of 35% - 50 % by 2030 - 2040 [3]. Wind is one of the cleanest and renewable sources of energy available. [4]. Horizontal axis wind turbines (HAWTs) represent one of the largest wind generators currently deployed worldwide, but commonly require strong and steady winds, are relatively large, and depend on wind direction, limiting their suitability for some applications [5][6]. Vertical Axis Wind Turbines (VAWTs), in contrast, can operate with variable wind conditions and offer advantages for urban environments, but their relatively low aerodynamic efficiency remains a major limitation [7][8]. Previous studies have explored biomimetic features such as tubercles on a humpback whale and hammerhead shark, and surface roughness modifications on golf-ball inspired dimples improve airflow, reduce drag, and enhance aerodynamic performance [9][10]; however, there remains a need for experimental validation of the combined biomimetic features and surface roughness technology and their effectiveness for VAWT applications [11].

To address this gap, this study aims to develop WinDTurbine Power, a hybrid dimple-tubercle blade to optimize the power generation of a VAWT utilizing NACA0012 airfoil model. The study combines Computational Fluid Dynamics (CFD) simulation, 3D printed blades, experimental wind tunnel validation, and wind speed field testing to evaluate aerodynamic performance, power coefficient (Cp), and energy generation across smooth (control), tubercle, dimple, and hybrid blade configurations.



Image1: Biomimetic tubercles' hammerhead shark [12]
Image 2: Roughness surface dimples' golf ball [13]

Research Questions

This study aims to develop hybrid dimple-tubercle blade configuration of a VAWT. It aims to answer the following questions:

1. To what extent do the CFD simulation and experimental validation for the hybrid blade differ relative to L/D and flow separation?
2. Does the hybrid blade (H-NACA0012) differ significantly from the smooth (S-NACA0012), tubercle (T-NACA0012), and dimple (D-NACA0012) blades in terms of L/D and flow separation?
3. What differences exist in mechanical and power output between hybrid and other blade configurations when tested ranging 3 to 10 m/s with a 5° angle?
4. At which wind speed does the hybrid VAWT reach its maximum Cp, and how close is it to the theoretical Betz limit (0.593)?

Literature Review

Previous studies have examined VAWTs as an alternative to conventional HAWTs. It demonstrates optimal performance in strong, steady winds but their substantial size, dependency on wind direction, and elevated installation costs present limitations for urban applications [4][5]. VAWTs offer omnidirectional operation and simplified installation, making them suitable for settings with variable wind conditions, although their comparatively low aerodynamic efficiency remains a major limitation [6][7]. The NACA 0012 airfoil is commonly used as a baseline for wind turbine aerodynamic research because of its symmetrical geometry and well-documented characteristics [14][15].

Researchers have explored biomimetic and surface roughness modifications to improve blade performance. Inspired by the flippers of humpback whales, tubercle structures have demonstrated improved power output and torque over traditional designs [4][16]. Similarly, the cephalofoil of hammerhead sharks provides a biological model for improving manoeuvrability and reducing hydrodynamic drag [17][18]. Golf-ball dimples have also been investigated as a surface modification, with dimple diameter, depth, position, and spacing influencing aerodynamic performance [10][19]. Notably, combining dimple and tubercle features on the Wortmann FX-60-126 airfoil enhanced lift generation while delaying drag onset, suggesting potential benefits from combining different surface modifications [20].

CFD enables controlled analysis of airflow behavior and serves as a virtual wind tunnel for evaluating aerodynamic designs [21]. Previous studies have examined airfoils angles 5°, 10°, and 20° and wind speeds 10, 20, and 30 m/s [22][23][24]. The efficiency with which a turbine transforms wind kinetic energy into usable mechanical power is quantified by the power coefficient (Cp) derived from CFD outputs. Turbine's performance conventionally assessed relative to the Betz Limit, a theoretical maximum efficiency of 59.3% that serves as a benchmark for evaluating turbine performance [25].

Overall, previous studies support the use of biomimetic modifications, surface roughness, CFD simulation, experimental wind tunnel testing, and wind speed field testing for improving wind turbine performance. However, their integration on a single VAWT blade remains unexplored. This study addresses that gap by the combined tubercle and dimple configuration on an NACA0012 VAWT blade through an integrated CFD simulation, experimental validation, and actual wind speed field testing.

Proposed Methodology: A quantitative experimental approach will be used in this study.

1. **Design and Model:** The blade designs for the four VAWT configurations will be created using Onshape CAD software and will be based on the NACA0012 symmetric airfoil. The smooth blade will be extruded to have a 290 mm span and a chord of 100 mm. The tubercled blade will also have leading-edge tubercles at different wavelengths (29 mm and 58 mm) and amplitudes (10% and 20% of the chord), inspired by the humpback whale's tubercles and hammerhead shark cephalofoils to decrease drag and increase the stability of flow. The dimpled blade will be designed with dimples on the surface that will be patterned after golf balls with four different placements covering 50% of the chord length, which increases the stability of airflow and lift. The hybrid blade blended the base form of tubercles with full hexagonal dimples to achieve maximum aerodynamic efficiency.

2. **CFD SimScale Simulation:** CFD simulations of the four blade configurations will be conducted using SimScale at 5 and 10 m/s at a 5° angle. Aerodynamic performance of each blade focuses on L/D ratio and wind flow separation.

3. **VAWT Blade 3D Print:** The blade designs with the best results will be prepared for 3D printing using Polylactic Acid (PLA) material. The printed blades will be used for the experimental wind tunnel validation and urban rooftop VAWT prototype.

4. **Wind Tunnel Test:** To validate CFD performance, the 3D printed blades will be tested at 5 and 10 m/s and a 5° angle, confirming their L/D ratio and other measured results.

5. **Fabrication and wind speed test:** The hybrid blade and the other blade configurations will be assembled independently with three straight blades into a H-rotor roof top VAWT. The three equally designed blades will be attached to the support, which will be connected to the rotating hub. The rotating hub will be positioned on top of the DC motor. The motor will be connected to the tower shaft, which will be fixed to a circular base to provide stability and reduce vibration during operation. The rotor hub will hold three equally spaced 3D PLA-printed blades, each shaped along a circular arc with a 254 mm inner radius and a 5° semi-arc pitch angle. This will be mounted on a 491 mm tall, 10 mm thick tower shaft, which in turn connects to a 320 mm steel round base. Wind speed testing will be conducted from 3 to 10 m/s. Rotational energy of the VAWT will be converted using a DC Motor 12-24v and 1200-24000RPM, with the positive and negative wires attached, which will allow the electrical power to flow through a capacitor to stabilize its electrical energy. Voltage, current, and rotational speed will be measured through repeated trials to determine the power output of each blade configuration.

6. **Power Coefficient (Cp), Scaled-Up Efficiency, and Betz Limit Calculation:** Using the measured power output, the power coefficient (Cp) will be calculated for each blade configuration to evaluate the scaled-up efficiency and compare the energy-generation performance of the four blade designs. The calculated Cp values will also be compared with the Betz Limit of 59.3% to determine the percentage of the theoretical maximum energy extraction achieved by each configuration.

Expected Outcome

The study is expected to demonstrate that the proposed hybrid dimple-tubercle VAWT blade can achieve improved aerodynamic efficiency and actual efficiency compared with the other blade configurations. The results are expected to validate the design through CFD simulation, wind tunnel testing, and wind speed field testing. Overall, the study aims to establish the hybrid design as a promising approach for improving VAWT wind energy generation.

Potential Limitation

The VAWT prototype is in a small-scale dimension with a 12-24 volts DC motor capable of minimal voltage generation.

Conclusion

WinDTurbinePower aims to develop and evaluate a hybrid dimple-tubercle VAWT blade for improved aerodynamic performance and energy conversion. Through CFD

simulation, experimental validation, and wind speed field testing, the study seeks to determine the potential of the hybrid design for more efficient small-scale wind energy generation.

Project Practicalities:

Figure 2 presents a Gantt chart outlining the schedule planned for this study.

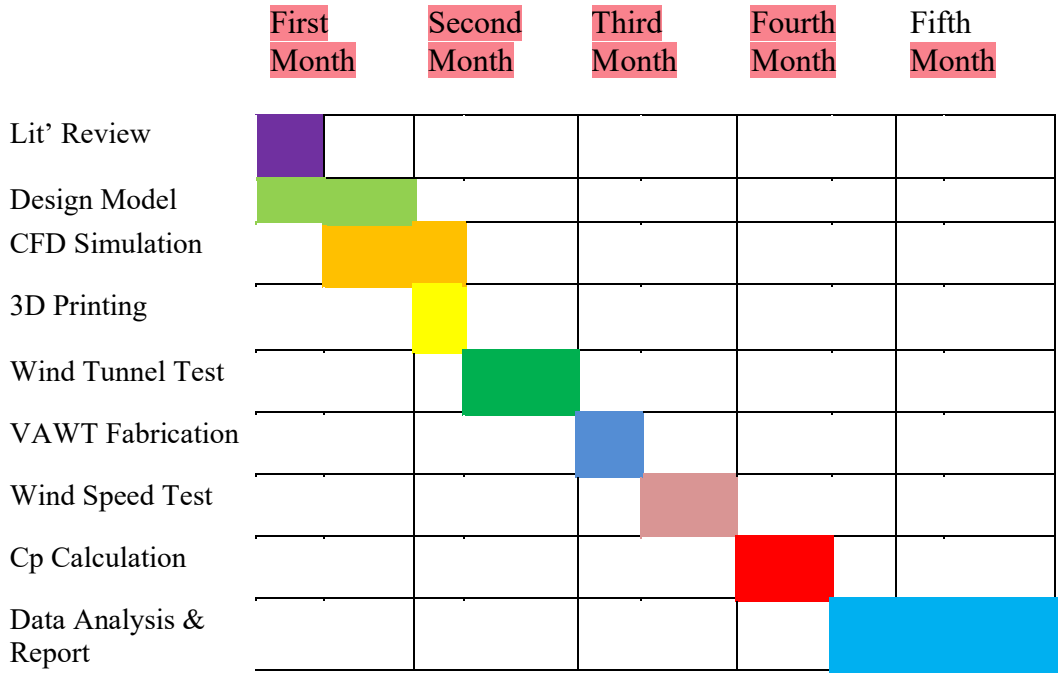


Figure 1. Timeline of the proposed study.

The project will span approximately 5 months. The first month will focus on the literature review, designing and modelling the VAWT blades, and conducting CFD simulations. The second month will involve 3D printing of blades and conducting wind tunnel testing. The third month will focus on VAWT fabrication, wind speed testing, and power coefficient calculations. Data analysis will be done on month 4, while the final month will focus on finalizing the research report.

Post-Program Plan

Upon completion, the researcher intends to pursue utility model protection for the proposed VAWT blade design and explore collaborations with the Department of Energy and local emerging startups industries. Building on the current design, the prototype will be refined for practical, small-scale wind energy generation in urban residential areas. Academic research conferences such as Science Castle Asia or submission to a relevant scientific journal may provide platforms for presenting the study and its results, while the researcher aims to promote VAWT as a potential renewable energy solution and encourage its wider use and further development globally.

REFERENCES

[1] E. Energy, "World surpasses 40% clean power as renewables see record rise," Ember-energy.org. Accessed: Sep. 19, 2026. [Online]. Available: https://ember-energy.org/latest-updates/world-surpasses-40-clean-power-as-renewables-see-record-rise/

[2] B. Lelis, "'Funding Gaps Hold Back Philippines RE Ambitions,'" Philstar.com. Accessed: Sep. 19, 2026. [Online]. Available: https://www.philstar.com/business/2025/01/22/2415880/funding-gaps-hold-back-philippines-re-ambitions

[3] V. Tachev, "Renewable Energy in the Philippines – Current State and Future Roadmap," Energy Tracker Asia. Accessed: Sep. 19, 2026. [Online]. Available: https://energytracker.asia/renewable-energy-in-the-philippines-current-state-and-future-roadmap/

[4] D. Rozehnal and J. Hnidka, "Performance Analysis of a Horizontal Axis Wind Turbine," MATEC Web of Conferences, vol. 313, p. 00052, 2020, doi: 10.1051/matecconf/202031300052.

[5] A. Dutta, H. Singh, G. K. Rai, and A. Bose, "Optimum Design of a Savonius VAWT for Implementation in Urban Places Having Low to Medium Wind Velocities," EPJ Web of Conferences, vol. 325, pp. 01020–01020, Jan. 2025, doi: 10.1051/epjconf/202532501020. |[6] S. Tumse, M. Bilgili, A. Yildirim, and B. Sahin, "Comparative Analysis of Global Onshore and Offshore Wind Energy Characteristics and Potentials," *Sustainability*, vol. 16, no. 15, pp. 6614–6614, Aug. 2024, doi: 10.3390/su16156614

[7] H. Seifi Davari, M. Seify Davari, R. M. Botez, and H. Chowdhury, "Advancements in Vertical Axis Wind Turbine Technologies: A Comprehensive Review," Arabian Journal for Science and Engineering, vol. 50, no. 4, pp. 2169–2216, Nov. 2024, doi: 10.1007/s13369-024-09723-x.

[8] S. Le Fouest and K. Mulleners, "Optimal Blade Pitch Control for Enhanced Vertical-Axis Wind Turbine Performance," Nature Communications, vol. 15, no. 1, p. 2770, Mar. 2024, doi: 10.1038/s41467-024-46988-0.

[9] P. Prakash, L. K. Pendyala, R. C. Teladevalapalli, and S. Mitra, "A Real-Time Experimental Investigation of Thin Cambered Tubercle Blades for Performance Improvement of Darrieus Vertical Axis Wind Turbine," pp. 257–261, Mar. 2023, doi: 10.1109/icgea57077.2023.10125996.

[10] H. Ali, M. R. Rasani, Z. Harun, and M. A. Shahid, "CFD Based Design Optimization of Dimples Induced on Blended Wing Body Airframe Using the Taguchi Method," PLOS ONE, vol. 20, no. 4, p. e0320885, Apr. 2025, doi: 10.1371/journal.pone.0320885.

[11] W. Yossri, S. Ben Ayed, and A. Abdelkefi, "Airfoil Type and Blade Size Effects on the Aerodynamic Performance of Small-Scale Wind Turbines: Computational Fluid Dynamics Investigation," Energy, vol. 229, p. 120739, Aug. 2021, doi: 10.1016/j.energy.2021.120739.

[12] E. Moulton, "Meet the Hammerheads: Discover All 10 Species and Places to Dive With Them," Diventures. Accessed: Sep. 19, 2026. [Online]. Available: https://www.diventures.com/meet-the-hammerheads-discover-all-10-species-and-places-to-dive-with-them/

[13] PNGTree, "Golf Ball 3d Element," Dec. 5, 2023. [Online]. Available: https://pngtree.com/freepng/golf-ball-3d-element_14621624.html

[14] M. Z. Iqbal, A. Al-Faruk, and Md. A. Islam, "Experimental Analysis of Aerodynamic Characteristics of NACA 0012 Wing Model With Multiple Winglets," Journal of Engineering Advancements, vol. 6, no. 1, pp. 22–30, Mar. 2025, doi: 10.38032/jea.2025.01.004. https://pngtree.com/freepng/golf-ball-3d-element_14621624.html

[15] S. Mitchell, I. Ogbonna, and K. Volkov, "Aerodynamic Characteristics of a Single Airfoil for Vertical Axis Wind Turbine Blades and Performance Prediction of Wind Turbines," Fluids, vol. 6, no. 7, p. 257, Jul. 2021, doi: 10.3390/fluids6070257.

[16] P. Prakash, L. K. Pendyala, and S. Mitra, "An Experimental Investigation of Novel Bio-Inspired Curved Bladed VAWTs Having Leading-Edge Tubercles for Enhanced Power Production," Sādhanā: Academy Proceedings in Engineering Sciences, vol. 49, no. 4, Oct. 2024, doi: 10.1007/s12046-024-02619-4.

[17] M. K. Gaylord, E. L. Blades, and G. R. Parsons, "A Hydrodynamics Assessment of the Hammerhead Shark Cephalofoil," Scientific Reports, vol. 10, no. 1, p. 14495, Sep. 2020, doi: 10.1038/s41598-020-71472-2.

[18] H. Song, Z. Ye, Y. Wang, and C. Li, "Enhancing Aerodynamic Performance of Vertical Axis Wind Turbines Using Bionic Airfoils Inspired by Swordfish Tail," Meccanica, vol. 59, no. 11, pp. 2059–2080, Sep. 2024, doi: 10.1007/s11012-024-01887-y.

[19] R. Allarton, J. Yao, T. Clifford, B. Hitchborn, L. J. Parker, and J. Shaw, "Surface Flow Modification of Aerofoils for Automotive Racing Car Applications," International Journal of Modern Physics B, vol. 34, no. 14–16, p. 2040096, Apr. 2020, doi: 10.1142/s0217979220400962.

[20] J. S. S. Manideep, L. C. Nixon, K. N. S. Ramu, and R. Eswaran, "Effects of Leading-Edge Tubercles and Dimples on a Cambered Airfoil and Its Performance," IRJET, Apr. 2022. Accessed: Sep. 19, 2026. [Online]. Available: https://www.irjet.net/archives/V9/i4/IRJET-V9I4195.pdf

[21] M. M. Elsakka, D. B. Ingham, L. Ma, and M. Pourkashanian, "Comparison of the Computational Fluid Dynamics Predictions of Vertical Axis Wind Turbine Performance Against Detailed Pressure Measurements," ResearchGate. Accessed: Sep. 19, 2026. [Online]. Available: https://www.researchgate.net/publication/350558009_Comparison_of_the_Computational_Fluid_Dynamics_Predictions_of_Vertical_Axis_Wind_Turbine_Performance_Against_Detailed_Pressure_Measurements

[22] V. Sadadiwala and D. Sharma, "A Detailed Study: CFD Analysis of NACA 0012 at Varying Angles of Attack," International Journal for Research in Applied Science and Engineering Technology, vol. 9, no. VII, pp. 2330–2336, Jul. 2021, doi: 10.22214/ijraset.2021.36843.

[23] S. Gupta, J. Zhao, A. Sharma, A. Agrawal, K. Hourigan, and M. C. Thompson, "Two- and Three-Dimensional Wake Transitions of a NACA0012 Airfoil," Journal of Fluid Mechanics, vol. 954, p. A26, Jan. 2023, doi: 10.1017/jfm.2022.958.

[24] A. A. Molaa and M. A. Abdulwahid, "Numerical and Experimental Study of the Impact on Aerodynamic Characteristics of the NACA0012 Airfoil," Open Engineering, vol. 14, no. 1, p. 20220506, Jan. 2024, doi: 10.1515/eng-2022-0506.

[25] A. Betz, "Das Maximum der theoretisch möglichen Ausnützung des Windes durch Windmotoren/ The Maximum of the Theoretically Possible Exploitation of Wind by Means of a Wind Motor," Scirp.org. Accessed: Sep. 19, 2026. [Online]. Available: https://www.scirp.org/reference/referencespapers?referenceid=1642440